

# Manifold Theory

PENG

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## 1 Set Theory

### 1.1 Relations

**Definition 1.1.** *Equivalence relation  $\sim$ :*

- *reflexive:*  $x \sim x$
- *symmetric:*  $x \sim y \Rightarrow y \sim x$
- *transitive:*  $x \sim y, y \sim z \Rightarrow x \sim z$

**Definition 1.2.** *Equivalence class of each  $x \in X$ :*

$$[x] = \{y \in X : y \sim x\}$$

**Definition 1.3.** *The collection of all  $[x]$  is  $X/\sim$*

**Example 1.4.** *Let  $A = \{1, 2, 3\}$ ,  $R = \{(x, y) : x = y\}$  on  $A$*

- *reflexive:*  $x = x$
- *symmetric:*  $x = y \Rightarrow y = x$
- *transitive:*  $x = y, y = z \Rightarrow x = z$

*Thus  $R$  is an equivalence relation.*

*Equivalence class:*

$$[1] = \{1\}$$

$$[2] = \{2\}$$

$$[3] = \{3\}$$

**Example 1.5.** *Let  $A = \{1, 2, 3\}$ ,  $R = \{(x, y) : x + y \text{ is even}\}$  on  $A$*

- *reflexive:*  $x + x \text{ is even}$
- *symmetric:*  $x + y \text{ is even} \Rightarrow y + x \text{ is even}$
- *transitive:*  $x + y \text{ is even}, y + z \text{ is even} \Rightarrow \begin{cases} x \text{ is even}, y \text{ is even}, z \text{ is even} \\ x \text{ is odd}, y \text{ is odd}, z \text{ is odd} \end{cases} \Rightarrow x + z \text{ is even}$

Thus  $R$  is an equivalence relation.

Equivalence class:

$$[1] = \{1, 3\}$$

$$[2] = \{2\}$$

$$[3] = \{1, 3\}$$

**Definition 1.6.** A partition of  $X$  is a collection of disjoint nonempty subsets of  $X$  whose union is  $X$ .

**Example 1.7.**  $X = \{1, 2, 3\}$

$$P = \{\{2\}, \{1, 3\}\}$$

$$P = \{\{1\}, \{3\}, \{2\}\}$$

$$P = \{\{1, 2, 3\}\}$$

$\{\{1\}, \{2\}, \{3\}\}$ : nonempty

$\{\{1\}, \{2\}\}$ : union is not  $X$

$\{\{1, 3\}, \{2, 3\}\}$ : disjoint

$\{\{1, 3\}, \{2, 4\}\}$ : union is not  $X$

**Exercise 1.8.** Collection of equivalence class  $\Rightarrow$  partition

Partition  $\Rightarrow$  unique equivalence relation

*Proof.*  $X/\sim \Rightarrow$  partition:

For  $\forall x_1, x_2 \in X$ :

$$[x_1] = \{y \in X : y \sim x_1\}$$

$$[x_2] = \{y \in X : y \sim x_2\}$$

If  $x_1 \not\sim x_2, \forall y \in [x_1] \Rightarrow y \notin [x_2] \Rightarrow$  disjoint nonempty subsets

If  $x_1 \sim x_2, \forall y \in [x_1] \Rightarrow y \in [x_2] \Rightarrow [x_1] = [x_2]$  □

*Proof.* Partition  $\Rightarrow \sim$  □

## 2 Euclidean Spaces

$$x = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} \in \mathbb{R}^n$$

**Definition 2.1.** For vectors:

$$x \cdot y = x^T y = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

**Definition 2.2.** Norm of  $x$ :

$$|x| = \sqrt{(x_1)^2 + (x_2)^2 + \cdots + (x_n)^2}$$

**Exercise 2.3.**

$$\max\{|x_1|, |x_2|, \dots, |x_n|\} \leq |x| \leq \sqrt{n} \max\{|x_1|, |x_2|, \dots, |x_n|\}$$

*Proof.* Let  $x_m = \max\{|x_1|, |x_2|, \dots, |x_n|\}$ :

$$\sqrt{(x_m)^2} \leq \sqrt{(x_1)^2 + \cdots + (x_m)^2 + \cdots + (x_n)^2} \leq \sqrt{(x_m)^2 + \cdots + (x_m)^2}$$

□

**Definition 2.4.**  $\theta$  between  $x$  and  $y$ :

## 3 Topological Spaces

Topology [1]

Four mechanisms (normal, segment, merge, torus) exist in every fluid point: fractal.

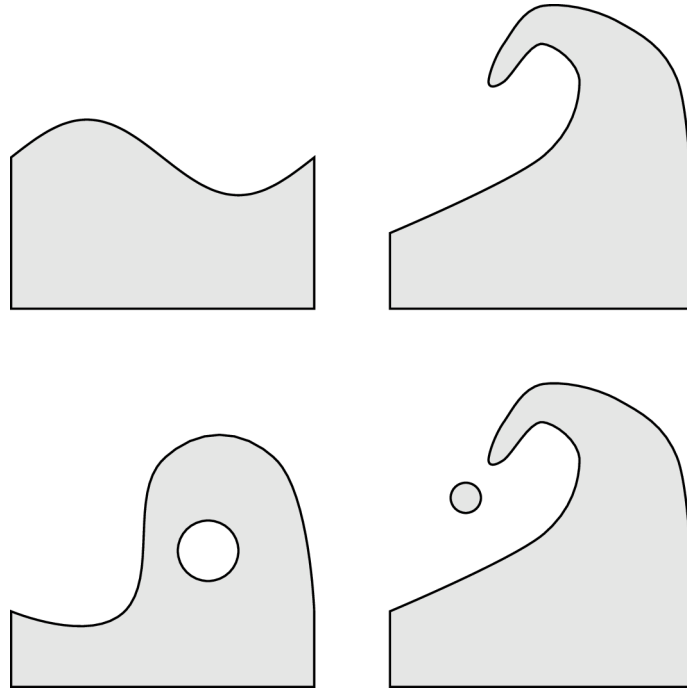


Figure 3.1: Sloshing topology.

## References

- [1] J.M. Lee. *Introduction to Topological Manifolds*. Springer New York, 2010. ISBN: 9781441979391.